

Work Instruction — BV-13 Assembly

Manufacturing Engineering · 2025-06 · Revision B

1. Scope

This instruction covers the assembly of the BV-13 valve assembly from received sub-components to the point of final inspection. It applies at all sites holding the assembly in scope of work.

Field	Entry
Instruction	WI-BV13
Revision	Revision B
Issued	2025-06
Applies to	BV-13 valve assembly

Operators shall work to the current revision only. Superseded revisions are withdrawn from the line and retained by Manufacturing Engineering.

2. Tooling

Item	Specification	Calibration
Torque wrench	Range covering the specified value	6 months
Assembly fixture	Fixture FX-BV13-01	Annual verification
Seal insertion tool	Tool ST-16	Visual check each shift
Depth gauge	Per drawing tolerance	6 months

The torque wrench shall be selected such that the specified value falls within the middle of the instrument range.

3. Sub-assembly

Components shall be drawn from released stock only and shall carry the batch identifier on the accompanying documentation. Clean all mating faces before assembly.

Step	Operation	Check
3.1	Clean mating faces	Visual
3.2	Inspect seal seating groove	Visual and gauge
3.3	Fit seal using insertion tool ST-16	Visual
3.4	Verify seal seating depth	Depth gauge

The seal shall be fitted using the insertion tool ST-16. The insertion tool for this assembly is not interchangeable with tooling for other assemblies in the series; confirm the tool number before fitting.

4. Housing assembly

The housing is secured to the valve body with the M10 housing bolts supplied in the assembly kit. Confirm that the kit is complete and that the fastener dimension matches this instruction, as kits for other assemblies in the series carry a different fastener and are not interchangeable.

Present the housing squarely and seat it by hand until the flange faces meet. Engage all fasteners by hand before applying torque. Fasteners shall be fitted dry; do not apply lubricant, sealant or thread locking compound.

Tighten in a diagonal sequence to 55 Nm. Recheck after 30 minutes.

Step	Operation	Record
4.1	Confirm kit complete and fastener dimension	Build record
4.2	Seat housing, confirm flange contact	Visual
4.3	Engage all fasteners by hand	Visual
4.4	Apply specified torque in sequence	Build record
4.5	Recheck at the stated interval	Build record

Where the recheck finds a fastener below the specified value, the sequence shall be reapplied in full and a further recheck performed.

5. Inspection

Check	Method	Acceptance
Flange contact	Visual and feeler	No measurable gap
Fastener torque record	Build record	All entries present
Recheck record	Build record	Present, at stated interval
External condition	Visual	Free of damage and residue

An assembly failing any check shall be quarantined and reported. Conforming assemblies shall be labelled and passed to packing.